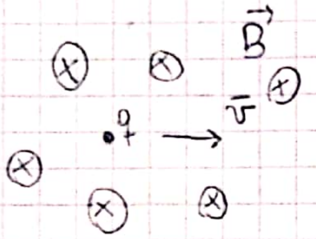
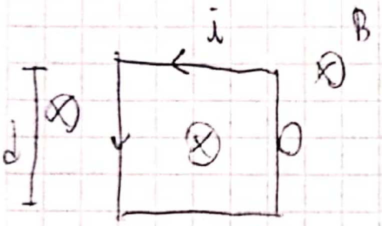
Magnetismo

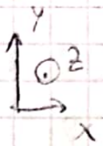
$q, \vec{v}, \vec{B}$

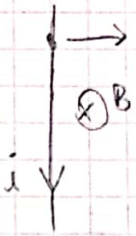
$$dF = i \cdot d\vec{l} \times \vec{B}$$



Ejercicio 7







$\longrightarrow F = i\,B\,dl\ \check{i}$

$\downarrow dl = -dl\ \check{j}$

$$\overline{dF} = i\ \overline{dl} \times \bar{B}$$
$$= i\,(-dl)\,\check{j} \times (-B)\,\check{k}$$
$$= +i\ dl\ B\ \underbrace{\check{j} \times \check{k}}_{\check{i}} \Rightarrow dF = i\,B\cdot dl$$

$$\Rightarrow \bar{F} = \int_d i\,B\cdot dl = i\,B\,D\ \check{i}$$

Entonces, para sacar velocidad

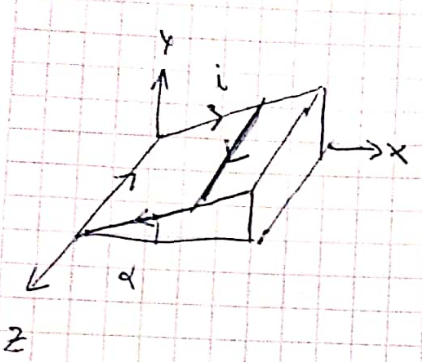
$$F = m\cdot \bar{a}$$

$$i B d = m \frac{dv}{dt} \Rightarrow \int_{t_0}^{t} \frac{iBd}{m}\, dt = \int_{v_0}^{v} dv$$

$$\boxed{\frac{iBd}{m}\, t = v(t)}$$ (parte del reposo)

en i versor

Ejercicio 10

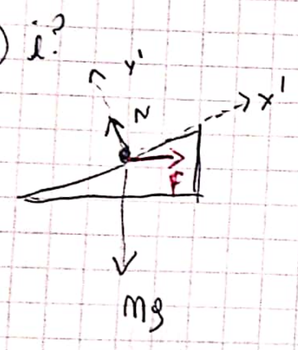


$\alpha = 37^\circ$

$\bar{B} = -0{,}2\,T\ \check{j}$

$L = 0{,}4\ m$

ⓐ ¿i?



$$dF = i\, d\bar{l} \times \bar{B}$$
$$= i\, dl\ \check{k} \times (-B)\ \check{j}$$
$$= -i\, B\, dl\ \ \check{k} \times \check{j}$$
$$= -i\, B\, dl\ (-\check{i})$$
$$= i\, B\, dl\ \check{i}$$

$$\Rightarrow \bar{F} = \int_L i B\, dl\ \check{i} \quad \Rightarrow \quad \bar{F} = i\, B\, L\ \check{i}$$

$$\sum F_{my'} = 0 \quad \Rightarrow \quad \bar{N} - \bar{P}_y - \bar{F}_y = 0$$

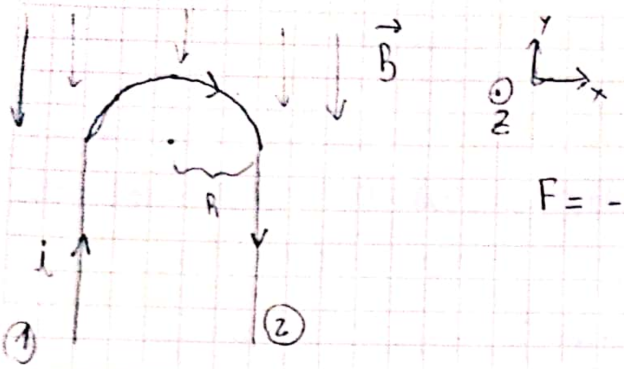
$$\sum F_{x'} = 0 \longrightarrow \quad F_{x'} - P_{x'} = 0$$

$$i\, B\, L \cos(\alpha) = m\, g\, \text{sen}(\alpha)$$

$$\Rightarrow \boxed{i = \frac{m\, g\ \text{sen}(\alpha)}{B\, L \cdot \cos(\alpha)}} \quad \Rightarrow \quad i = 2{,}8125\ A$$

NOTA

Ejercicio 8



$F = -2iBR\,\check{k} \rightarrow$ demostrar

$dl = R\,d\theta$

- $\overline{dl_1} \parallel \overline{B} \Rightarrow \overline{F_1} = 0$
- $dl_2 \parallel \overline{B} \Rightarrow \overline{F_2} = 0$
- $dF_3 = i \cdot \overline{dl} \times \overline{B}$

$\overline{F_3} = \int i \cdot [dl_x\,\check{i} + \underbrace{dl_y\,\check{j}]\times(-B)\,\check{j}}_{\bar{0}}$

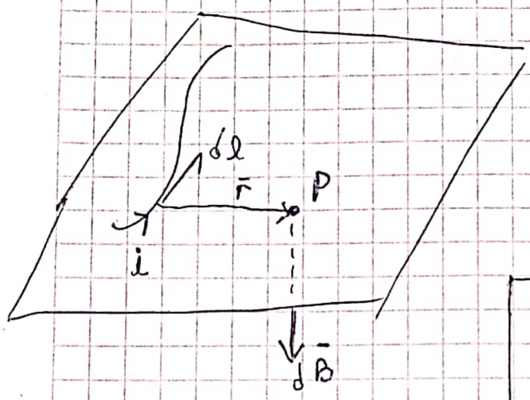
$\int i \cdot dl \cdot \cos(\theta)\,\check{i} \times (-B)\,\check{j}$

$\int -i\,dl\cos(\theta)\,B\,\check{k}$

$-iBR\int_{-\pi/2}^{\pi/2}\cos(\theta)\,R\,d\theta\,\hat{k} = -iB\,\mathrm{Sen}(\theta)\Big|_{-\pi/2}^{\pi/2}\,\check{k}$

$= -iBR(2)\,\check{k} = \boxed{-2iBR\,\check{k}}$

Ley de Biot - Savart



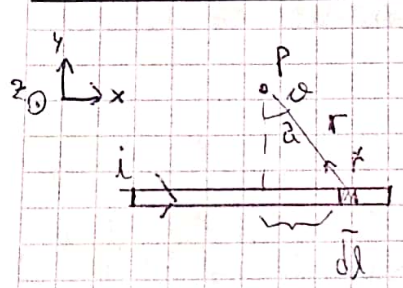
$\bar{dB} \perp \bar{dl}$

$\bar{dB} \perp \bar{r}\ (\check{r})$ } en simultaneo

$$\boxed{\bar{dB} = \frac{\mu_0}{4\pi}\, i\, dl \times \frac{\hat{r}}{r^2}}$$

$\mu_0 = 4\pi \cdot 10^{-7}\ \frac{Tm}{A}$

Campo $\vec{B}$ de un Conductor Rectilineo



$$\vec{dB} = \frac{\mu_0}{4\pi}\, i\, dl \times \frac{\check{r}}{r^2}$$

$$\vec{dB} = \frac{\mu_0}{4\pi}\, i\, dx\, \hat{i} \times \frac{(-x\hat{i} + a\hat{j})}{(x^2+a^2)^{3/2}}$$

$$= \frac{\mu_0}{4\pi}\, \frac{i\, dx\, a}{(x^2+a^2)^{3/2}} (\hat{i} \times \hat{j})$$

$$= \frac{\mu_0}{4\pi}\, \frac{i\, dx}{(x^2+a^2)^{3/2}}\, \hat{z}$$

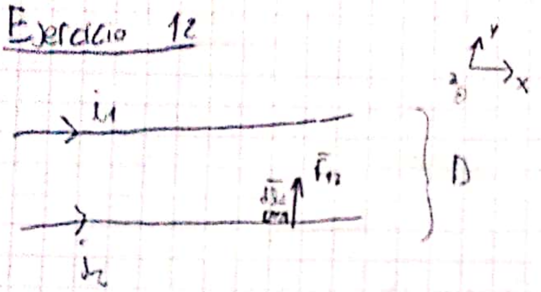
$$\Rightarrow \bar{dB} = \frac{\mu_0}{4\pi}\, \frac{i}{a} \cos(\theta)\, d\theta\, \hat{k}$$

$$\bar{B} = \int \frac{\mu_0}{4\pi}\, \frac{i}{a} \cos(\theta)\, d\theta\, \hat{k} = \boxed{\frac{\mu_0 i}{4\pi a} \left(\operatorname{sen}(\theta_2) - \operatorname{sen}(\theta_1) \right) \hat{k}}$$

- Si el hilo es $\infty \Rightarrow \boxed{\bar{B} = \frac{\mu_0 i}{2\pi a}\, \hat{k}}$

NOTA

Ejercicio 12



$i_1 = 0,2\,A$ $\qquad L = \infty$

$i_2 = 0,3\,A$

$D = 0,6\,m$

- $B_1 = \dfrac{\mu_0\, i_1}{2\pi r}$; $B_2 = \dfrac{\mu_0\, i_2}{2\pi r}$

ⓐ $\dfrac{F_{12}}{l}$?

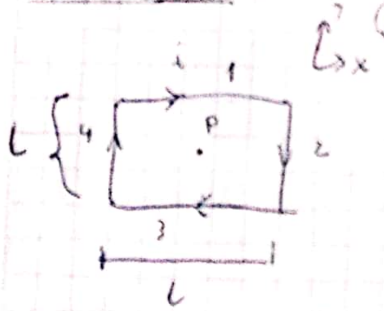
$$\overline{dF_{12}} = i_2\,\overline{dl_2} \times \overline{B_1} = i_2\,dl_2\,\hat{i} \times (-B_1)\,\hat{k}$$

$$= -i_2\,dl_2\,\underbrace{\hat{i}\times\hat{k}}_{-\hat{j}} = i_2 \cdot dl_2\, B_1\,\hat{j}$$

$$= i_2\,dl_2\,\frac{\mu_0\, i_1}{2\pi r}\,\hat{j} = i_2\,dl_2\,\frac{\mu_0\, i_1}{2\pi D}\,\hat{j}$$

$$\Rightarrow \bar{F}_{12} = \int i_1\, i_2\,\frac{\mu_0}{2\pi D}\,dl\,\hat{j} \Rightarrow \boxed{\frac{F_{12}}{L} = i_1\, i_2\,\frac{\mu_0}{2\pi D}\,\hat{j}} = 2\cdot 10^{-8}\,\frac{N}{m}\,\hat{j}$$

Ejercicio 18



ⓐ

$$\vec{B}_1 = \frac{-\mu_0\, i}{4\pi\,\frac{L}{2}}\,\mathrm{sen}(\theta_2) - \mathrm{sen}(\theta_1)\;\hat{k}$$

$$B = \frac{-\mu_0\, i}{4\pi\,\frac{L}{2}}\left[\mathrm{sen}(\pi/4) - \mathrm{sen}(-\pi/4)\right]\hat{k}$$

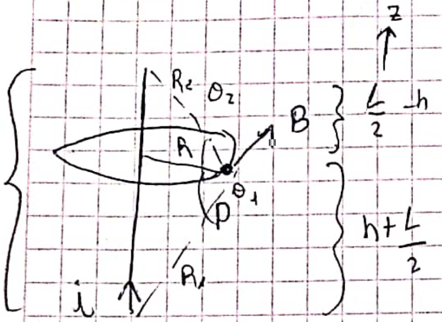
$$= -\frac{\mu_0\, i}{4\pi\, L/2}\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) = \boxed{-\frac{\mu_0\, i}{2\pi L}\cdot\sqrt{2}\;\hat{k}}$$

$\Rightarrow$ Por superposición

$$\boxed{B_T = 4\,B_1 = \frac{-\mu_0\, i\cdot 2\,\sqrt{2}}{\pi L}\,\hat{k}}$$

Escaneado con CamScanner

Ejercicio 16



(a) P:

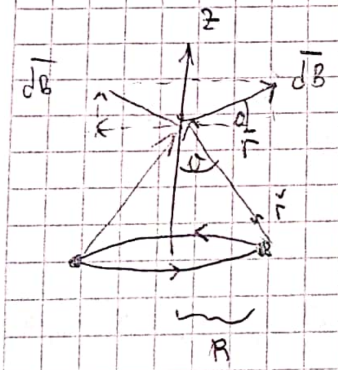
$$B = \frac{\mu_0 i}{4\pi} \left[\text{Sen}(\theta_2) - \text{Sen}(\theta_1) \right]$$

$$R_2 = \sqrt{R^2 + \left(\frac{L}{2}-h\right)^2}$$

$$R_1 = \sqrt{R^2 + \left(h+\frac{L}{2}\right)^2}$$

$$B = \frac{\mu_0 i}{4\pi} \left[\frac{\frac{L}{2}-h}{\sqrt{R^2+\left(\frac{L}{2}-h\right)^2}} + \frac{h+\frac{L}{2}}{\sqrt{R^2+\left(h+\frac{L}{2}\right)^2}} \right]$$

Campo $\bar{B}$ generado por una espira



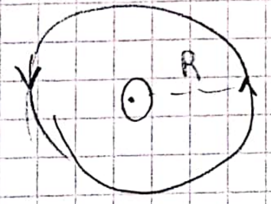
$$dB = \frac{\mu_0}{4\pi} i \frac{\bar{dl} \times \check{r}}{r^2} \quad \Big| \; dl = R\, d\theta$$

$$B(0,0,z) = B_z \check{k}$$

$$= \int_l dB_z \check{k}$$

$$= \int_l \frac{\mu_0}{4\pi} \frac{i R^2 d\theta}{(R^2+z^2)^{3/2}}$$

$$\boxed{\Rightarrow \bar{B}_{(0,0,z)} = \frac{\mu_0 i R^2}{2 (R^2+z^2)^{3/2}}}$$



$$\boxed{B(0,0,0) = \frac{\mu_0 i}{2R} \check{k}}$$

$$\boxed{B(\phi) = \frac{\mu_0 i}{4\pi} \frac{\phi}{R}}$$ $\rightarrow$ Para Tramos de espira

NOTA

CLASE 20

HOJA N°

FECHA

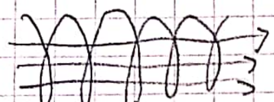
Inductancia

$\Phi \propto B \propto I$

$\Phi = L I$ → Inductancia

$\frac{\Phi}{I} = L$ → N, l, S ; μ (μ_0)

$[L] = \frac{W}{A} = H$ (Henry) → Flujo por unidad de corriente



N, S, l

$$\Phi = \iint \vec{B}\, N\, d\vec{S} = \mu_0 \frac{N}{l} I N \cdot S$$

$$\Phi = \frac{\mu_0 N^2 S}{l} I$$

$$\Rightarrow \frac{\Phi}{I} = \frac{\mu_0 N^2 S}{l} \Rightarrow \boxed{L = \mu_0 m^2 l S}$$

$m = \frac{N}{l}$

- **Solenoide ideal** ($l \to \infty$)

$\frac{L}{l} = \mu_0 m^2 S$ → Se calcula por unidad de longitud.

- **Toroide ideal**

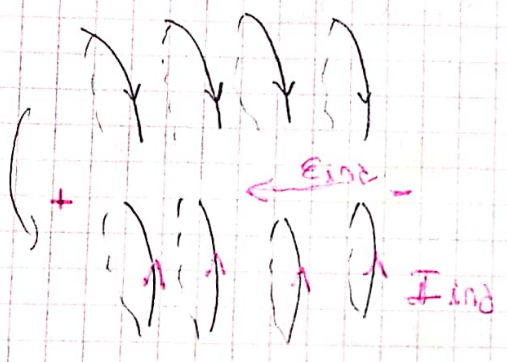
$$L = \frac{\mu_0 N^2 S}{2\pi R}$$

$S \to \pi R_E^2$

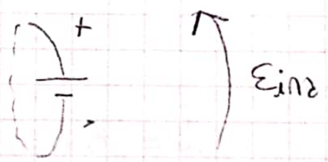
Ley de Faraday - Lenz

$$\mathcal{E}_{ind} = -\frac{d\Phi}{dt} = -L\frac{dI}{dt}$$

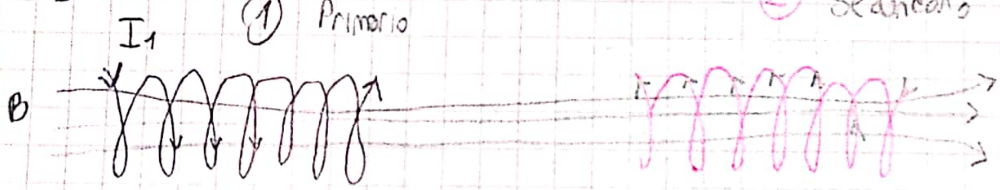
Si hay un flujo variable, va a aparecer una fem que va a contrarrestar la variación del flujo



$\uparrow I(t) \Rightarrow \uparrow \Phi(t)$



Ej:



$$\Phi_1 = L_1 I_1$$

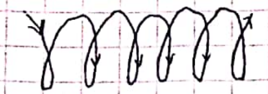
$$\Phi_{21} = M I_1 \qquad [M] = H$$

$$\frac{d\Phi_{21}}{dt} = M\frac{dI_1}{dt} \Rightarrow \mathcal{E}_2 = -\frac{d\Phi_{21}}{dt} = -M\frac{dI_1}{dt}$$

HOJA Nº

FECHA

Ejercicio 16



$\ell = 0{,}4\,m \qquad r = 0{,}02\,m$

$N = 2000$

$I(t) = 2A \ \mathrm{sen}(\pi t)$

ⓐ $L = \mu_0 \dfrac{N^2}{\ell} \pi r^2 = \boxed{15{,}8 \ mH_v}$

ⓑ $B(t) \Rightarrow \Phi(t)$

$$\varepsilon_{ind} = \frac{d\Phi}{dt} = L \frac{di}{dt} = L \cdot 2A \cdot 100\pi \cos(100\pi t)$$
$$= 200\pi \ A \ L \cos(100\pi t)$$
$$= 200\pi A \ 15{,}8\,mH \ \cos(100\pi t)$$
$$= 3160\pi \ mVolt \ \cos(100\pi t)$$
$$\boxed{\varepsilon_{ind} = 3{,}16\,\pi \ Volt \ \cos(100\pi t)}$$

ⓒ 2do bobina dentro

¿M, ε_{ind_2}?

$N_2 = 100$

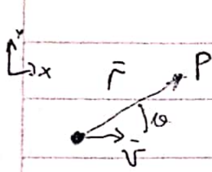
$r_2 = 0{,}5\,cm$

$$\Phi_{21} = \iint B_1 N_2 \, dS_2 = B_1 N_2 S_2 = B_1 N_2 \pi r_2^2$$
$$\mu_0 \frac{N_1}{l_1} i(t) \ N_2 \pi r_2^2$$

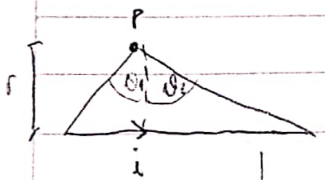
$$M \left\{ \frac{\Phi_{21}}{i(t)} = \boxed{\mu_0 \frac{N_1}{l_1} N_2 \pi r_2^2 = M} \right.$$

Resumen Campos $\vec{B}$

$$d\bar{B} = \frac{\mu_0}{4\pi}\, i\, \bar{dl} \times \frac{\hat{r}}{r^2} \Rightarrow \bar{B} = \int_l d\bar{B}$$

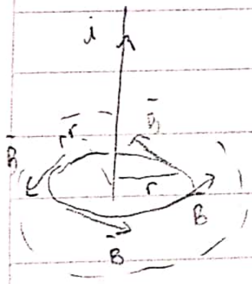


$$\bar{B} = \frac{\mu_0}{4\pi}\, q \cdot \frac{\bar{v} \times \hat{r}}{r^2} = \frac{\mu_0}{4\pi}\, \frac{q \cdot v\, \text{sen}(\theta)}{r^2}\, \hat{k}$$

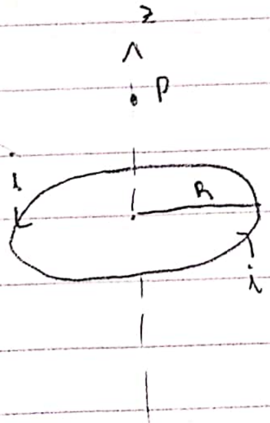


$$\vec{B} = \frac{\mu_0\, i}{4\pi\, r} \left[\text{sen}(\theta_2) - \text{sen}(\theta_1)\right] \hat{k}$$

$$\xrightarrow[\text{centro}]{\text{en el}} B = \frac{\mu_0\, i}{2\pi\, r}\, \text{sen}(\theta)\, \hat{k}$$

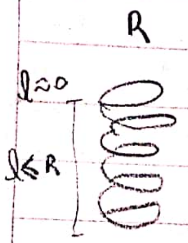


$$\bar{B} = \frac{\mu_0\, i}{2\pi r}\, \hat{\phi} \xrightarrow{\text{es: } r' = 2r} B = 2B'$$

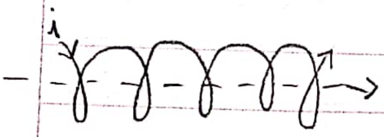


$$B(0,0,z) = \frac{\mu_0\, i\, R^2}{2\left(R^2 + z^2\right)^{3/2}}\, \hat{k}$$

$$B(0,0,0) = \frac{\mu_0\, i}{2R}\, \hat{k}$$



$$B = N \frac{\mu_0 i}{2R} \hat{k}$$

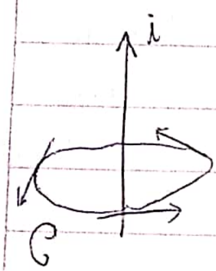


$L = \infty \hookrightarrow B = \mu_0 \, n \, i \, \hat{i}$] Solenoide ideal

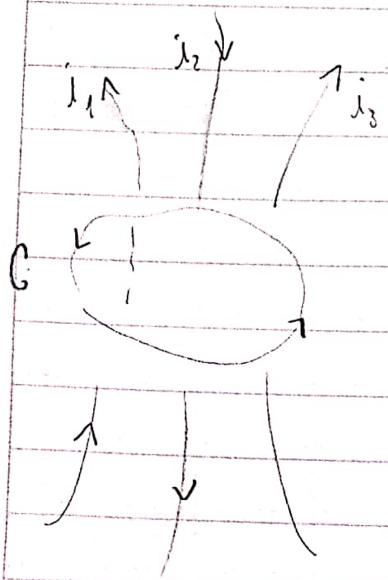
$N, L \left(n = \frac{N}{L} \right)$

$$B = \frac{\mu_0 N i}{2\pi R} \hat{\phi}$$] TOROIDE IDEAL

Ley de Ampere

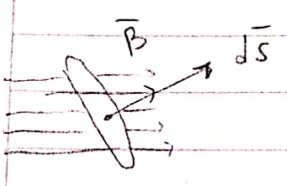


$$\boxed{\oint_C \bar{B} \cdot d\bar{l} = \mu_0 \, i_{conc.}}$$



$i_C = i_1 - i_2 + i_3$

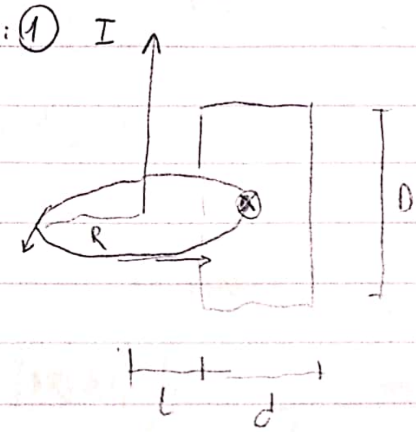
Flujo Magnético



$$\Phi = \iint_S \bar{B} \cdot d\bar{S}$$

$$[\Phi] = T \cdot m^2 = Weber\ (W)$$

Ej: 1



$l = 0{,}01\,m$

$d = 0{,}02\,m$

$D = 0{,}05\,m$

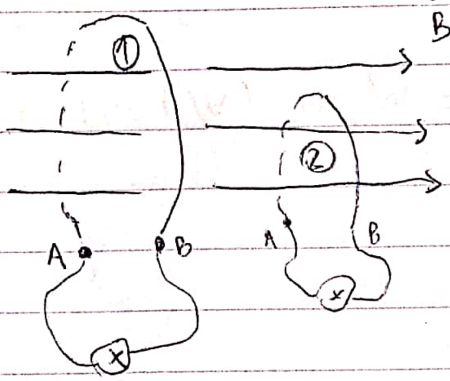
$$d\Phi = \bar{B} \cdot d\bar{S} \Rightarrow \Phi = \iint_S \bar{B} \cdot d\bar{S}$$

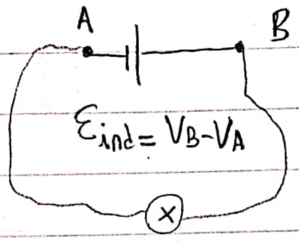
$$= \iint \frac{\mu_0 I}{2\pi r}\, dr\, dz = \frac{\mu_0 I}{2\pi} \cdot \int_0^D dz \int_l^{l+d} \frac{dr}{r}$$

$$\Rightarrow \frac{\mu_0 I Z}{2\pi} \cdot \ln\left(\frac{l+d}{l}\right) = \boxed{1{,}0986 \cdot 10^{-7}\ W}$$

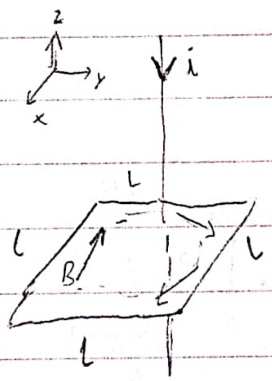
Ley de Faraday - Lenz

$$\frac{d\Phi}{dt} = -\varepsilon_{ind} = V_B - V_A$$





Ej: 3



$$B = \frac{\mu_0 I}{2\pi r}\hat{\phi}$$

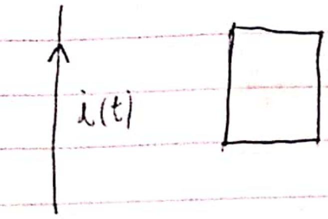
$$\bar{dS} = ds\,\hat{n}$$

$$\varepsilon_{ind} = \frac{-d\Phi}{dt}$$

$\Phi = \iint_S \bar{B} \cdot d\bar{S} = 0$ cte por $\perp$

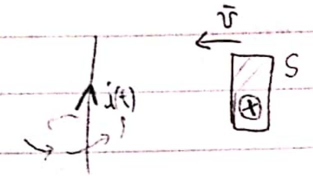
↳ no varía en el Tiempo

$$\Rightarrow \varepsilon_{ind} = \frac{-d\Phi}{dt} = \boxed{0}$$

⑥ 

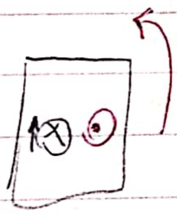
indicar dirección de corriente inducida

ⓐ la bobina se acerca al cable



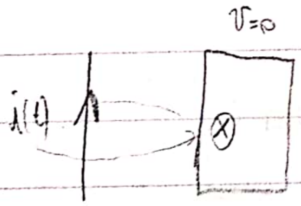
$\Phi = \iint_S \bar{B}\, d\bar{s}$; $B = \dfrac{\mu_0 I(t)}{2\pi r}$

• $\uparrow I(t)$; $\downarrow r \Rightarrow \uparrow \bar{B} \Rightarrow \uparrow \Phi \Rightarrow$ la fem inducida se opone a la variación de flujo

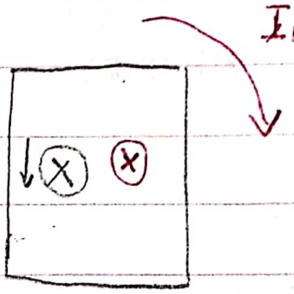


I_{ind} → Antihorario

ⓑ la bobina está quieta

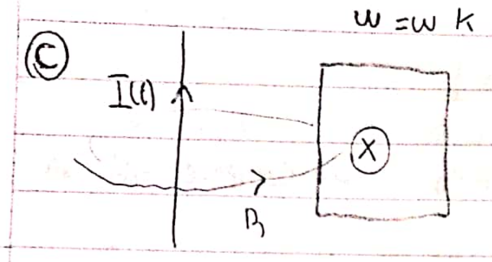


• $\downarrow I(t) \Rightarrow \downarrow B \Rightarrow \downarrow \Phi$



I_{ind} → horario

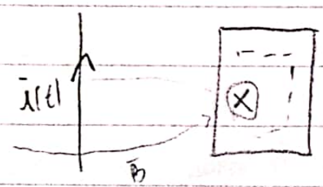
c)

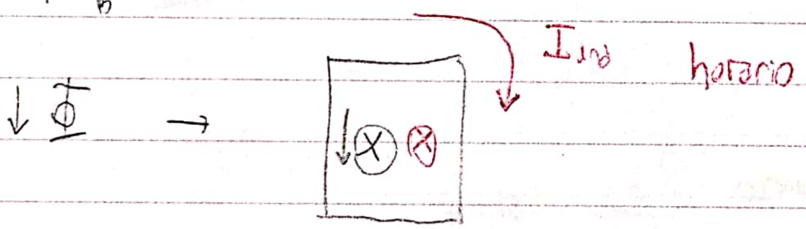


la bobina giro alrededor del campo

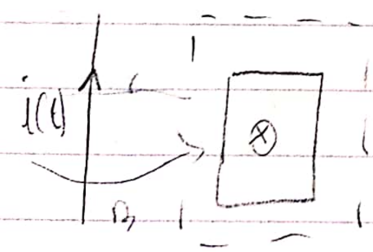
$\Phi = cte \Rightarrow \frac{d\Phi}{dt} = 0 \Rightarrow$ No hay Fem inducida

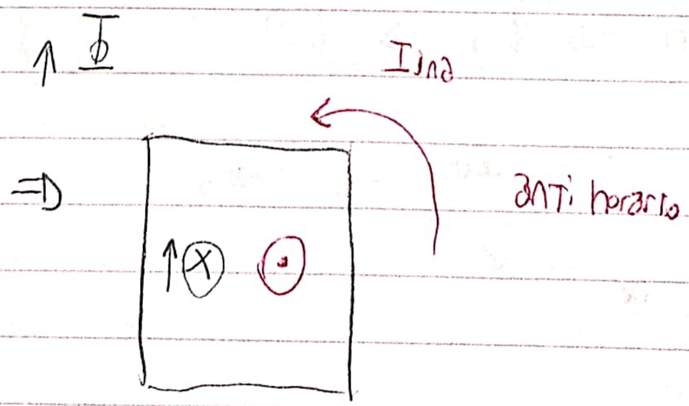
d) la bobina se achica y la corriente es constante





e) la bobina se agranda y la corriente es constante

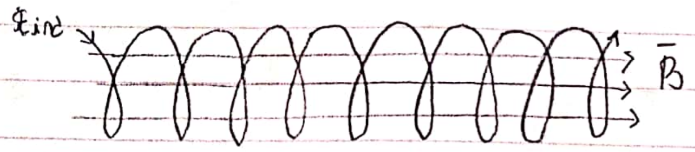




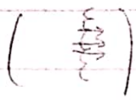
5) $N = 20$ $\qquad$ $R = 5\ \Omega$

$r = 1\ cm$

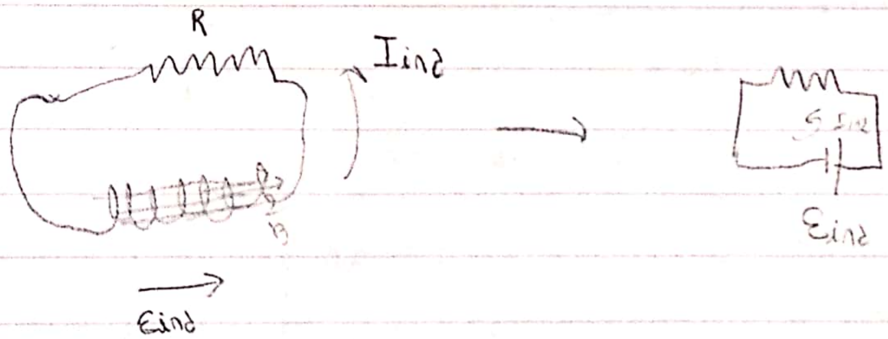
$B = 0{,}8\ T$



$$\Phi_{max} = B N \pi r^2 \quad ; \quad \Phi_{min} = 0 \rightarrow$$



$$\varepsilon_{ind} = -\frac{\Delta \Phi}{\Delta t} = -\frac{(\Phi_{min} - \Phi_{max})}{\Delta t} = -\frac{0 - BN\pi r^2}{\Delta t} = \frac{BN\pi r^2}{\Delta t}$$

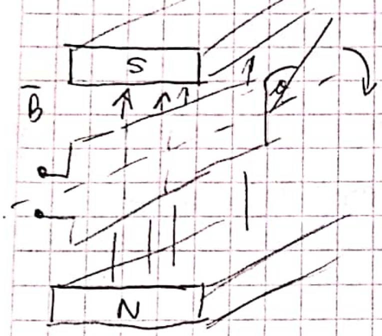


$$\varepsilon_{ind} = R\, I_{ind} \Rightarrow \frac{BN\pi r^2}{\cancel{\Delta t}} = R \cdot \frac{\Delta Q}{\cancel{\Delta t}} \Rightarrow \boxed{\Delta Q = \frac{BN\pi r^2}{R}}$$

HOJA N°

FECHA

Circuitos de Corriente Alterna



Eje $\vec{w}$

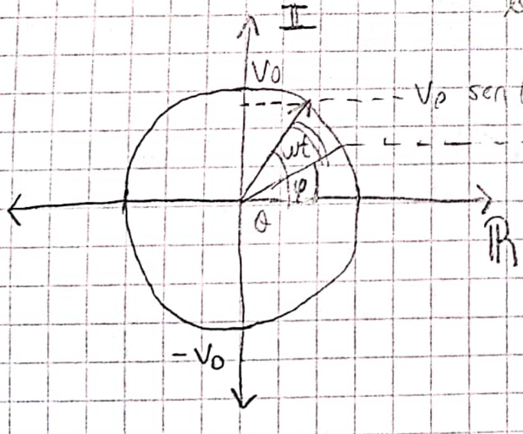
$\theta(t) = wt + \varphi$

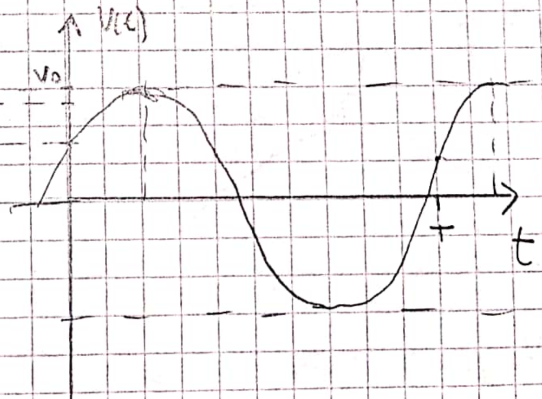
- $\Phi(t) = \iint_S \vec{B}\, d\vec{s} = \iint_S B\, ds \cos(\theta(t)) = B \cdot S \cos(\theta(t))$

- $\mathcal{E}(t) = -\frac{d\Phi}{dt}$ (Faraday - Lenz)

$= -BS\,(-w\,\text{sen}(wt+\varphi)) = \underbrace{BSw}_{\boxed{V_0}}\,\text{sen}(wt+\varphi)$

$\Rightarrow \mathcal{E}(t) = V_0 \cdot \text{sen}(\underbrace{wt+\varphi}_{\theta(t)})$



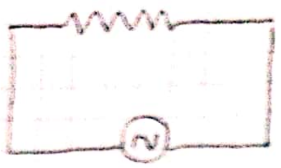


$A = V_0 \rightarrow$ Tensión máxima o de Pico

$W = 2\pi f \rightarrow$ Frecuencia angular o Pulsación

$f \rightarrow$ Frecuencia

Circuito Resistivo Puro



$E(t) = V_0 \operatorname{sen}(\omega t + \varphi)$

$E(t) = V_R$

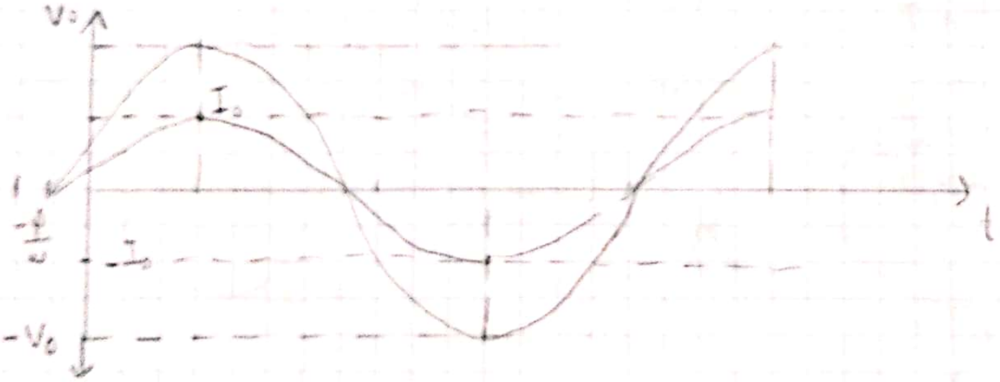
$V(t) = R \cdot I(t)$

$$I(t) = \frac{V(t)}{R} = \overbrace{\frac{V_0}{R}}^{I_0} \cdot \operatorname{sen}(\omega t + \varphi) = \boxed{I_0 \operatorname{sen}(\omega t + \varphi) = I(t)}$$

La corriente y el voltaje están en Fase

$$P(t) = I(t) \cdot V(t) = I_0 \operatorname{sen}(\omega t + \varphi) \cdot V_0 \operatorname{sen}(\omega t + \varphi)$$

$$\boxed{P(t) = \underbrace{I_0 V_0}_{I_0^2 R} \operatorname{sen}^2(\omega t + \varphi)}$$

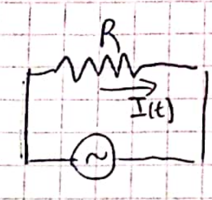


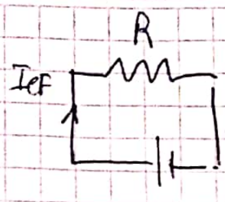
$$\boxed{P_M = \text{Pot media} = \frac{P_0}{2}}$$

→ Potencia activa o Pot media

HOJA N°

FECHA





$V_{eF} = R\,I_{eF} = R \cdot \frac{I_0}{\sqrt{2}}$

$Energ = P_M \cdot T$ ↳ Periodo

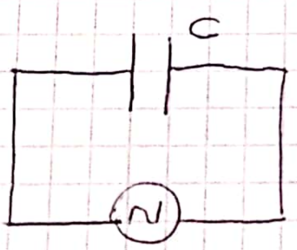
$= \frac{I_0 V_0}{2} T = \frac{I_0^2 R}{2} \cancel{T}$

$E = P \cdot T$

$= I_{eF}^2 R \cancel{T}$

$\boxed{I_{eF} = \frac{I_0}{\sqrt{2}}}$; $\boxed{V_{eF} = \frac{V_0}{\sqrt{2}}}$

Circuito capacitivo puro



$V(t) = V_0 \operatorname{sen}(\omega t + \varphi)$

$V(t) = V_c(t)$

$\frac{dQ(t)}{dt} = I(t)$

$V_0 \operatorname{sen}(\omega t + \varphi) = \frac{Q(t)}{C}$

$\to V_0\, \omega \cos(\omega t) = \frac{1}{C} I(t)$

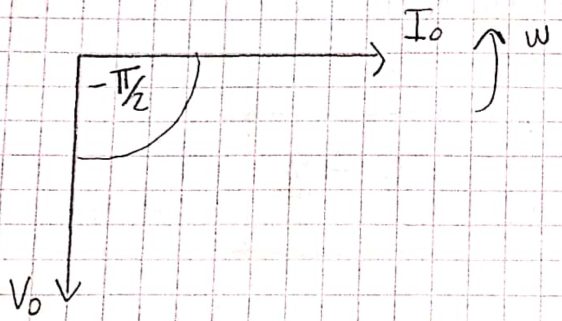
$\to \frac{V_0}{I_0} = \frac{1}{\omega C} = X_c$ → Reactancia capacitiva

↳ ¿Cuánto se opone el capacitor al paso de la corriente?

$\underline{Z} = R + j\,X \rightarrow$ Impedancia ~ Resist en C.C

$\hookrightarrow X_L - X_C$

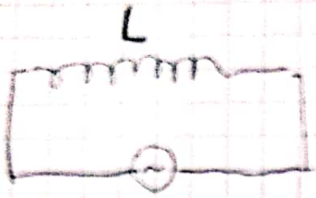
$V_0 = Z \cdot I_0$ ~ Ohm en C.C



$$P(t) = -\frac{V_0 I_0}{2}\,\text{sen}(2\omega t)$$

$$\boxed{P_A = P_M = 0}$$

Circuito Inductivo Puro



$I(t) = I_0 \, sen(\omega t)$

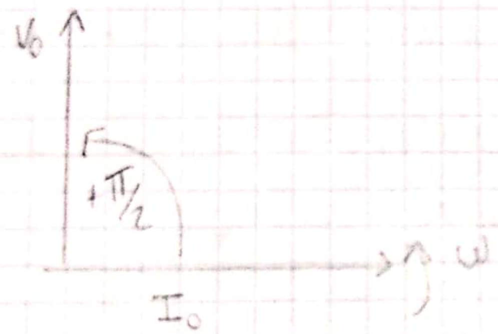
$+\pi/2 \to$ el voltaje adelanta $\pi/2$ a la corriente

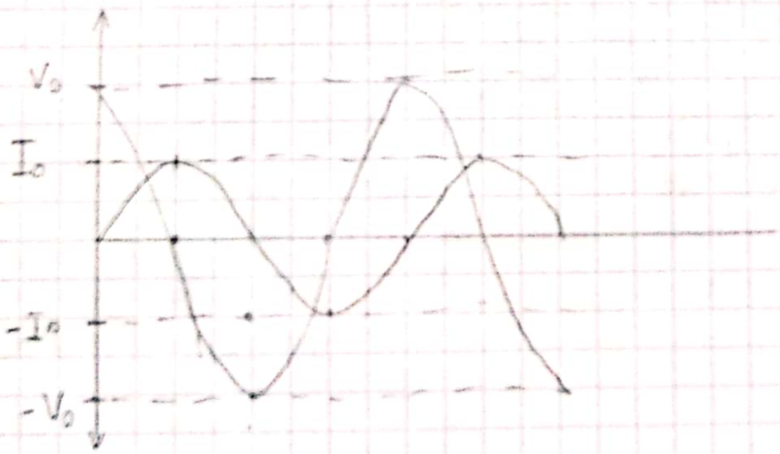
$V(t) = V_0 \, sen(\omega t + \varphi)$

$V(t) + V_L(t) = 0$ Por Kirchoff

$V(t) - L\frac{dI}{dt} = 0 \Rightarrow V(t) = L\frac{dI}{dt}$

- $\frac{V_0}{I_0} = \omega L = X_L \to$ Reactancia Inductiva





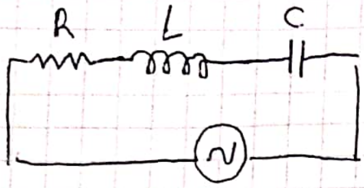
$V(t) = sen(\omega t + \pi/2)$
$= cos(\omega t)$

$P(t) = V(t)\,I(t) = \frac{V_0 I_0}{2}\, sen(2\omega t)$

$P(t) = V_{ef} \cdot I_{ef} \quad sen(\omega t 2)$

$P_A = P_m = 0$

Circuito RLC Serie



$$\mathcal{E}(t) = V(t) = V_0 \operatorname{sen}(\omega t + \varphi)$$

$$I(t) = I_0 \operatorname{sen}(\omega t)$$

⊙ $V(t) = V_R(t) + V_L(t) + V_C(t)$ → Kirchoff

$$\frac{dQ(t)}{dt} = I(t)$$

$$V_0 \operatorname{sen}(\omega t + \varphi) = R \cdot I(t) + L \frac{dI(t)}{dt} + \frac{Q(t)}{C}$$ → derivo m.a.m.

$$\cancel{\omega} V_0 \cos(\omega t + \varphi) = R I_0 \cancel{\omega} \cos(\omega t) - L I_0 \omega^{\cancel{2}} \operatorname{sen}(\omega t) + I_0 \operatorname{sen}(\omega t) \frac{1}{C\omega}$$

$$V_0 \cos(\omega t)\cos(\varphi) - V_0 \operatorname{sen}(\omega t)\operatorname{sen}(\varphi) = R I_0 \cos(\omega t) - L I \omega \operatorname{sen}(\omega t) + \frac{I_0}{\omega C} \operatorname{sen}(\omega t)$$

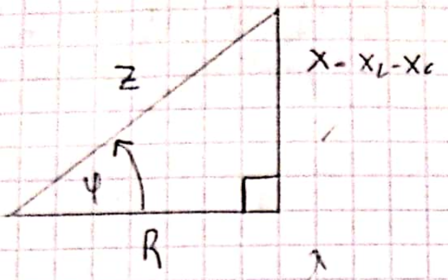
$$\left[V_0 \cos(\varphi) - R I_0\right] \cos(\omega t) = \left[V_0 \operatorname{sen}(\varphi) - I_0 \omega L + \frac{I_0}{\omega C}\right] \operatorname{sen}(\omega t)$$

$$\Rightarrow \cos(\varphi) = \frac{R I_0}{V_0} = \frac{R}{\underbrace{\frac{V_0}{I_0}}_{Z}}$$ → impedancia

$$\Rightarrow \operatorname{sen}(\varphi) = \left(\omega L - \frac{1}{C\omega}\right)\frac{I_0}{V_0} = \frac{\overbrace{\omega L}^{X_L} - \overbrace{\frac{1}{C\omega}}^{X_C}}{\underbrace{\frac{V_0}{I_0}}_{Z}} = \frac{\overbrace{X_L - X_C}^{X}}{Z} = \frac{X}{Z}$$

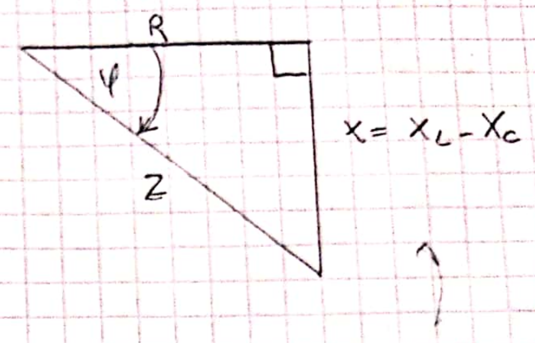
impedancia

CASO 1 $(X_L > X_C)$



CIRCUITO REACTIVO INDUCTIVO

CASO 2 $(X_L < X_C)$



CIRCUITO REACTIVO CAPACITIVO

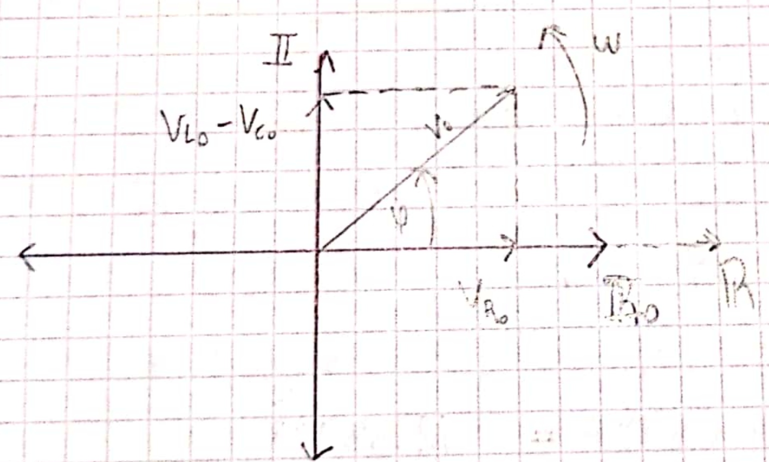
$$Z = \sqrt{R^2 + X^2}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$[Z] = \Omega$$

$$Z = R + j\,\underbrace{X}_{X_L - X_C}$$

Diagrama Fasorial



$$V_{eF} = \frac{V_0}{\sqrt{2}}$$

$$I_{eF} = \frac{I_0}{\sqrt{2}}$$

- $V_0 = Z \cdot I_0$
- $V_{R_0} = R \cdot I_0$
- $V_{L_0} = X_L \cdot I_0$
- $V_{C_0} = X_C \, I_0$

Ejemplo 1:

$V_0 = 100V$ $\quad L = 0,5\ Hr$

$F = 50\ Hz$ $\quad C = 4\ mF$

$R = 20\ \Omega$

ⓐ Z

ⓑ I_0, I_{ef}

ⓒ $V_{R_0}, V_{L_0}, V_{C_0}$, Vectores

ⓓ D.F ⓔ φ ⓕ f_0

ⓐ ¿Z? $\quad Z = \frac{V_0}{I_0}$

$$Z^2 = R^2 + (X_L - X_C)^2$$

$$= (20\,\Omega)^2 + \left(\omega L - \frac{1}{\omega C}\right) \qquad \omega = 2\pi F$$

$$= 400\,\Omega + \left(2\pi\, 50Hz - \frac{1}{2\pi 50Hz\ 4\cdot 10^{-3}F}\right)^2 =$$

$$400\,\Omega^2 + (\underbrace{157,075\ \Omega}_{X_L} - \underbrace{0,795\ \Omega}_{X_C})^2 \Rightarrow X_L > X_C \rightarrow \text{C. Reactivo inductivo}$$

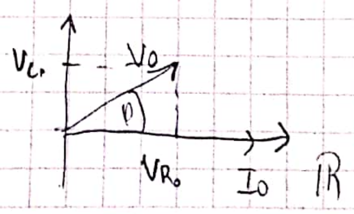
$$\Rightarrow Z^2 = 24823,12\ \Omega^2 \Rightarrow \boxed{Z = 157,55\ \Omega}$$

ⓑ $I_0 = \frac{V_0}{Z} = \frac{100\,V}{157,55\,\Omega} = \boxed{0,634\ A}$

$I_{ef} = \frac{I_0}{\sqrt{2}} = \frac{0,634\ A}{\sqrt{2}} = 0,4483\ A$

ⓒ
- $V_{R_0} = R \cdot I_0 = 20\,\Omega \cdot 0,634\,A = 12,68\ V$
- $V_{L_0} = X_L \cdot I_0 = 157,075 \cdot 0,634\,A = 99,58\ V$
- $V_{C_0} = X_C \cdot I_0 = 0,795\,\Omega \cdot 0,634\,A = 0,504\ V$
- $V_{ef} = \frac{V_0}{\sqrt{2}} =$

ⓓ



Voltaje adelanta la corriente

NOTA

HOJA N°

FECHA

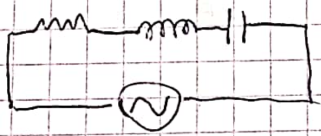
(e) ¿φ? $\quad \varphi = \text{arc.Tg}\left(\frac{X}{R}\right) = 82{,}7^{\circ}$

(F) Frec. de Resonancia $\rightarrow X_L = X_C$

$$\Rightarrow 2\pi F L = \frac{1}{2\pi F C} \Rightarrow F^2 = \frac{1}{(2\pi)^2 CL} \Rightarrow f = \sqrt{\frac{1}{(2\pi)^2 CL}}$$

$$\boxed{F_0 = 3{,}558 \text{ Hz}}$$

<u>Potencia de RLC serie</u>



$$\begin{cases} V(t) = V_0 \operatorname{sen}(\omega t + \varphi) \\ I(t) = I_0 \operatorname{sen}(\omega t) \end{cases}$$

$P(t) = I(t) \cdot V(t) \longrightarrow$ desarrollo

$$P(t) = \underbrace{\frac{I_0 V_0}{2} \cos(\varphi)}_{\text{P. activa}} - \underbrace{\frac{I_0 V_0}{2} \cos(2\omega t + \varphi)}_{\text{P. Fluctuante}}$$

Factor de Potencia

$$P_{ACT} = P_M = \frac{I_0}{2} \overbrace{V_0 \cos(\varphi)}^{V_{R_0}} = \frac{I_0 V_{R_0}}{2} = \frac{I_0}{\sqrt{2}} \frac{V_{R_0}}{\sqrt{2}} = I_{ef} V_{Ref}$$

$[P_A] = W$

$[P_R] = V.A.R$

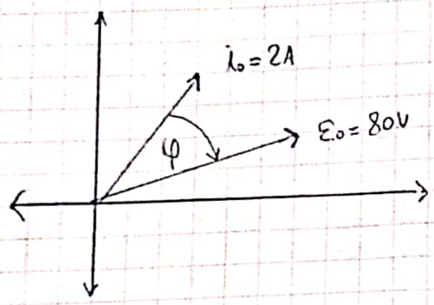
NOTA

Ejercicio 4

$P = 40\,W$

$F = 50\,Hz$

2 elementos pasivos



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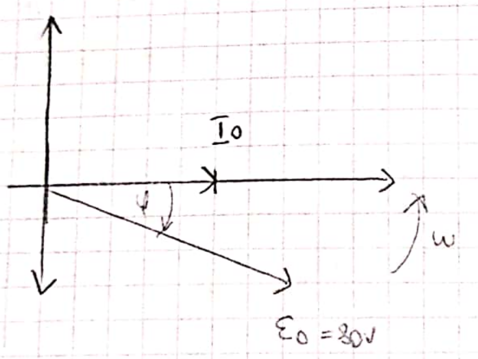


¿R, L, C?

→ corriente adelanta voltaje

⇒ ~~Inductivo~~ capacitivo

⇒ RC

$$\cdot\ P_{activo} = \frac{I_0 \cdot V_0}{2} \cos(\varphi) = \frac{I_0 \cdot V_{R_0}}{2} \qquad (1)$$

$$\cdot\ Z^2 = R^2 + (X_L - X_C)^2 = R^2 + X_C^2 = R^2 + \frac{1}{(2\pi F C)^2}$$

$$Z = \frac{V_0}{I_0} = \frac{80\,V}{2A} = 40\,\Omega \Rightarrow (40\,\Omega)^2 = R^2 + \frac{1}{(2\pi F C)^2} \qquad (2)$$

$$\xrightarrow{(1)} \quad 40\,W = \frac{I_0\, R \cdot I_0}{2} \Rightarrow R = \frac{40W \cdot 2}{(2A)^2} = \boxed{20\,\Omega}$$

$$\xrightarrow{en\ (2)} \quad 1600\,\Omega^2 = 400\,\Omega^2 + \frac{1}{C^2\, 4\pi^2} \Rightarrow \boxed{C = 9{,}19 \cdot 10^{-5}\ F}$$

$C = 91{,}19\ \mu F$

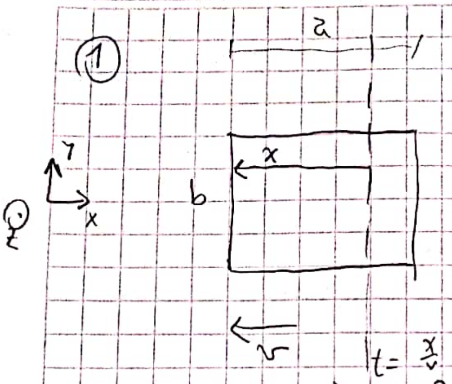
NOTA

PARCIAL

HOJA Nº

FECHA

①



$\otimes \bar{B}$

$a = 3cm$

$b = 2cm$

$\bar{v} = 1 m/s$

$B = 3000 \frac{W}{m^2} = 3000 T$

$t = \frac{x}{v}$

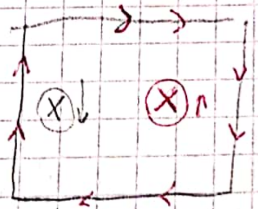
ⓐ $x = vt \Rightarrow \phi$?

$$\phi = \iint_S B \cdot ds = -B \int_0^{0,02} \int_0^{0,03} -dx = B \cdot (a - vt) \cdot b$$

$$= 3000 T \left(0,03m - \frac{1m}{s} t\right) \cdot 0,02m$$

$$= \boxed{1,8 \frac{W}{m} - 60 \frac{W}{m\,s} t}$$

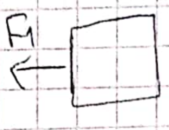
ⓑ $R = 6 \Omega \Rightarrow I$?



$$\varepsilon_i = \frac{-d\phi}{dt} = -(-60) = 60 V$$

$$\Rightarrow I(t) = \frac{V}{R} = \frac{60V}{6\Omega} = 10A$$

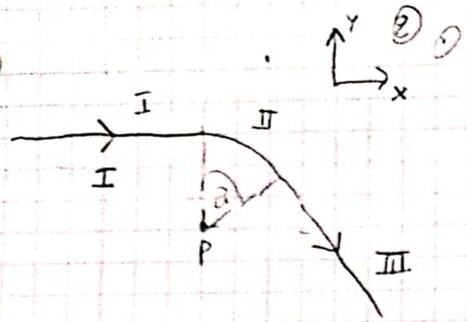
ⓒ W que debe realizar una fuerza sobre la espira para extraer complet. la espira de la región manteniendo $\bar{v}$ cte



$$\Rightarrow \bar{F}_1 = \int_0^b I \cdot dl \times B = I \cdot B \int_0^b dl = 10 \cdot 3000 \cdot 0,02$$

$$= 600 N$$

$$W = \vec{F} \cdot \vec{d} = 600^N \cdot 0,03^m = 18 J$$

② 

$\alpha = 60°$

R

I

$d\bar{l} = R \cdot d\bar{\theta}$

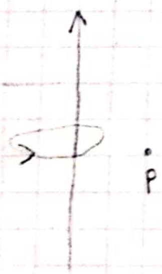
ⓐ B_{II} x B-S

$\Rightarrow \bar{dB} = \frac{\mu_0}{4\pi} \cdot i \cdot \frac{\bar{dl} \times \hat{r}}{r^2}$ $r = R$

$\bar{B} = -\frac{\mu_0 i}{4\pi R^2} \hat{z} \int_0^{\pi/3} d\varphi = -\frac{\mu_0 i}{4\pi R^{\cancel{2}}} \underbrace{\frac{\pi}{3}}_{60°}, \cancel{R} = \boxed{-\frac{\mu_0 i}{12 R} \hat{k} = \bar{B}}$

ⓑ $B_{I,II}$ x L. Ampere

$\oint_C B \cdot dl = \mu_0 \cdot I_{conc.}$



$B \cdot \oint dl = \mu_0 i$

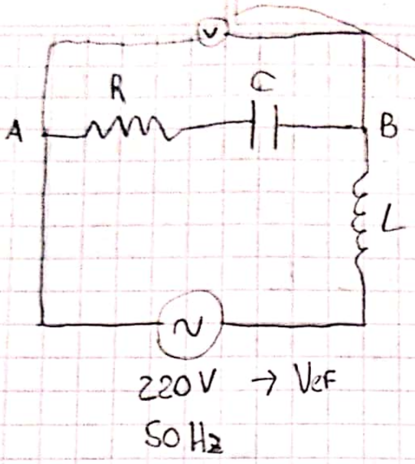
$B \cdot 2\pi r = \mu_0 i$

$B = \frac{\mu_0 i}{2\pi R}$ → Al ser semiinfinito y al estar en el extremo del cable, el campo será la mitad del generado por un cable infinito

$\Rightarrow \frac{1}{2} \cdot \frac{\mu_0 i}{2\pi R} = \boxed{\frac{\mu_0 i}{4\pi R}}$

NOTA

3



$V_{ef} = 155 \text{ V}$

$C = 500\ \mu F = 500 \cdot 10^{-6} F$

$R = 16\ \Omega$

$X_L = \omega L$

$X_C = \frac{1}{\omega C} = \frac{1}{2\pi F C} = 6{,}366\ \Omega$

a) Ieficaz en el circuito

$|R + X_C| = \sqrt{16^2 + 6{,}366^2} = 17{,}22\ \Omega$

$V_{ef\,A\text{-}B} = 155 V \Rightarrow I_{ef} = \frac{V_{ef}}{17{,}22\ \Omega} = \frac{155 V}{17{,}22\ \Omega} = 9 A$

b) ¿L?

$V_0 = V_{ef} \cdot \sqrt{2} = 311{,}13 V$

$I_0 = 9A \cdot \sqrt{2} = 12{,}73 A$

$Z = \frac{V_0}{I_0} = \frac{311{,}13}{12{,}73 A} = 24{,}44\ \Omega = \sqrt{R^2 + (X_L - X_C)^2}$

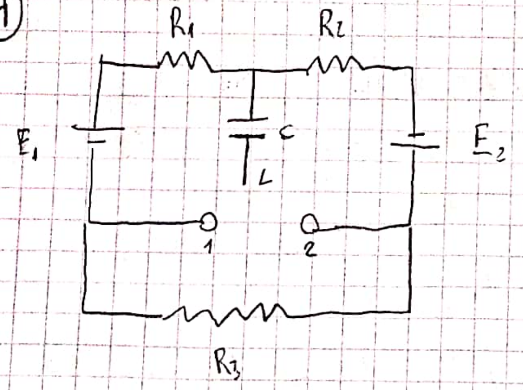
$\Rightarrow (24{,}44)^2\ \Omega^2 = (16\ \Omega)^2 + (X_L - X_C)^2$

$341\ \Omega^2 = X_L^2 - 2X_C X_L + X_C^2$

$300{,}47 = X_L^2 - 12{,}732\ X_L$ → cuadrática

$X_L = 24{,}83 = \omega L = 2\pi F L \Rightarrow L = \frac{24{,}83\ \Omega}{2\pi\ 50 Hz} = \boxed{0{,}079\ Hy = L}$

(4)



$R_1 = 400\,\Omega$

$R_2 = 200\,\Omega$

$R_3 = 400\,\Omega$

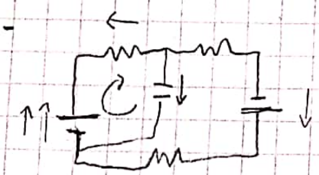
$E_1 = 10\,V$

$E_2 = 20\,V$

$C = 600\,\mu F$

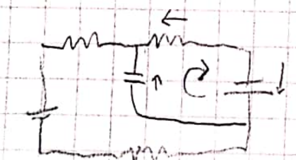
(a) Carga del capacitor con llave en 1 y en 2

1.-



- $R_T = R_1 + R_2 + R_3 = 1000\,\Omega$
- $I = \frac{V}{R} = \frac{10V + 20V}{1000\,\Omega} = 0{,}03\,A$
- $V_{R_1} = 0{,}03A \cdot 400\,\Omega = 12\,V$
- $V_C + V_{E_1} - V_{R_1} = 0 \Rightarrow V_C = 2V \Rightarrow Q_1 = V_C \cdot C = 2V \cdot 600 \cdot 10^{-6} = \boxed{1{,}2mC}$

2.-



$R_T = 1000\,\Omega$

$I = \frac{30V}{1000} = 0{,}03\,A$

- $V_{R_2} = 0{,}03A \cdot 200\,\Omega = 6\,V$

$V_{E_2} + V_C - V_{R_2} = 0$

$V_C = +6V - 20V$

$V_C = -14\,V \rightarrow$ invertir sentido

NOTA

$\Rightarrow Q_1 = V_C \cdot C = 14V \cdot 600 \cdot 10^{-6} = \boxed{8{,}4\,mC = Q}$

HOJA N°

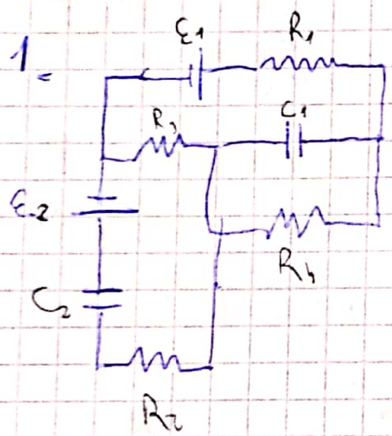
FECHA

(b) t que tarda el capacitor en cambiar el nuevo valor de carga al cambiar de 1 a 2.

$t \cong 5 \tau$?

PARCIAL 2

1.



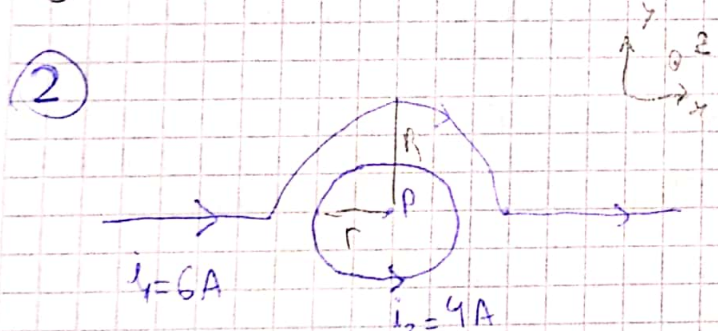
$\varepsilon_1 = 36V$ $\varepsilon_2 = 5V$

$R_1 = 3\Omega$ $R_2 = 5\Omega$ $R_4 = 2\Omega$

$C_1 = 250\ \mu F$, $C_2 = 125\ \mu F$

(a) Pot que disipa R_3

(2)



$r = 0{,}04m$

$R = 0{,}06m$

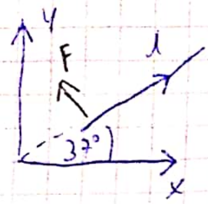
$\bar{B}$ en P?

$$\bar{B}_2 (0,0,0) = \frac{\mu_0 i}{2r}\hat{k} = 6{,}28\ 10^{-5}\ T\ \hat{k}$$

$$\bar{B}_1 = \frac{\mu_0 i}{4\pi}\frac{\theta}{R} - \hat{k} = \frac{4\pi\, 10^{-7}\, 6A}{4\pi}\frac{\cancel{180}\ \pi}{0{,}06} = -3{,}14 \cdot 10^{-5}\hat{k}$$

$$\Rightarrow \bar{B}_T = \bar{B}_1 + \bar{B}_2 = 6{,}28 \cdot 10^{-5} - 3{,}14 \cdot 10^{-5}\ \hat{k} = 3{,}14 \cdot 10^{-5}\hat{k}$$

③



⊗ B

$B = -2{,}5\,T\ \hat{k}$

$l = 80\ cm = 0{,}8\ m$

$i = 20\ A$

$dF = i \cdot d\bar{l} \times \bar{B}$

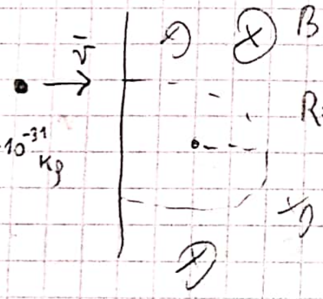
$F = \bar{i}\,\bar{l} \times \bar{B}$

$F = i \cdot l \cdot B \cdot \operatorname{sen} \overset{90}{\alpha} = 20\,A \cdot 0{,}8\,m \cdot 2{,}5\,T = 40\ N$

$$\bar{F} = i \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ l\cos\alpha & l\operatorname{sen}\alpha & 0 \\ 0 & 0 & -2{,}5 \end{vmatrix} = i \cdot [l\cos 37 \cdot (-2{,}5)\hat{i} - l\operatorname{sen} 37 \cdot (-2{,}5)\hat{j}]$$

$$= i\,[-1{,}2\ \hat{i} - 1{,}6\ \hat{j}]$$

④



$m_e = 9{,}1 \cdot 10^{-31}\,kg$

$R = 1{,}7\ mm = 0{,}17\ cm = 0{,}0017\ m$

$B = 0{,}002\ T$

¿v?

$\Sigma F = m a$

$F_m = m\, a_c$

$F = m \cdot \dfrac{v^2}{R}$

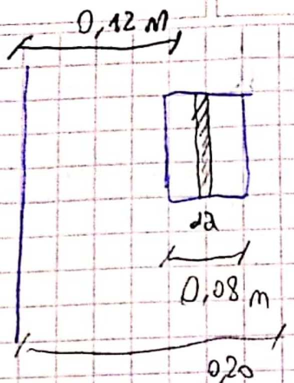
$|q|\,v\,B = m\,\dfrac{v^2}{R} \Rightarrow v = \dfrac{q\,B\,R}{m} = 597{,}8 \cdot 10^3\,\dfrac{m}{s}$

NOTA

HOJA N°

FECHA

(5)



$N = 40$

$i(t) = 4A \ \text{sen}(50 \ s^{-1} \ t)$

(a) expresión de fem en la bobina

$\bullet \ \mathcal{E}_{ind} = \frac{-d\Phi}{dt}$

$\bullet \ \Phi = \iint \bar{B} \cdot d\bar{S} \rightarrow B(t) = \frac{\mu_0 N I(t)}{2\pi a} =$

$= \frac{\mu_0 N i(t)}{2\pi} \cdot 0{,}15m \cdot \int_{0{,}12}^{0{,}20} \frac{da}{a}$

$= \frac{N\mu_0 i(t)}{2\pi} \cdot 0{,}15m \ \ln\left(\frac{0{,}20}{0{,}12}\right) = \frac{\cancel{4\pi} \cdot 10^{-7}}{\cancel{2\pi}} \cdot 0{,}15m \cdot 0{,}51$

$\varnothing_B = 1{,}53 \cdot 10^{-8} \ i(t) \cdot N = 6{,}12 \cdot 10^{-7} \ i(t)$

$\Rightarrow \mathcal{E}_{ind} = \frac{-d\varnothing}{dt} = 6{,}12 \cdot 10^{-7} \cdot 4A \cdot 50 s^{-1} \cos(50 s^{-1} t)$

$\Rightarrow \boxed{\mathcal{E}_{ind} = 1{,}224 \cdot 10^{-4} \cos(50 s^{-1} t)}$ ✓

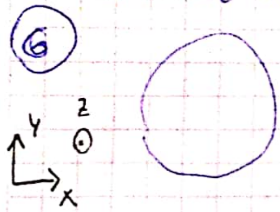
(b) inducción mutua M

$M = \frac{\varnothing_{B_{TOTAL}}}{i(t)} = \frac{6{,}12 \cdot 10^{-7} \ \cancel{i(t)}}{\cancel{i(t)}} = \boxed{6{,}12 \cdot 10^{-7} \ Hy}$

Escaneado con CamScanner

Pl XY

⑥



N espiras

$$B = \left(-4T + 0{,}5 \frac{T}{s} t\right) \hat{k}$$

sent de circ. antes y desp de $t=8$

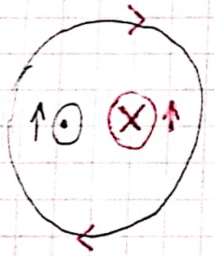
• $t<8$

B será entrante al plano pero estará disminuyendo su flujo

⇒ Para compensar, la fem inducida se opone a la variación de flujo

⇒ el sentido será horario

• $t>8$ → B será saliente al plano XY, y sigue incrementando su flujo



⇒ Para compensar, la fem ind. se opone a la variación de flujo

⇒ el sentido será horario

⑦ $R = 30\,\Omega$

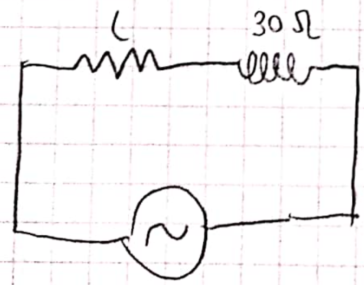
$\omega = 500\ s^{-1}$

$V_{ef} = 200\ V$

$V_{ef_R} = 160\ V$

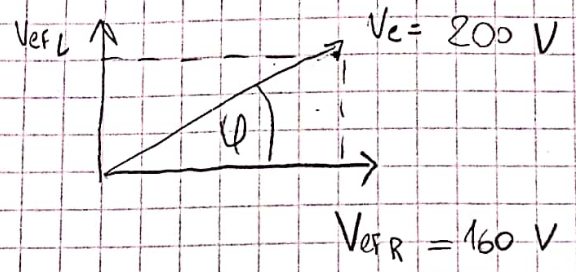
a) desfasaje y factor de potencia

ⓑ Autoinducción de la bobina



HOJA N°

FECHA



Son CAnTos

$$\cos(\varphi) = \frac{V_{ef_R}}{V_{ef}} = \frac{160\ V}{200} = 0{,}8$$

$$\Rightarrow \boxed{\varphi = 36{,}87^\circ} \rightarrow \text{desfasaje}$$

$$\boxed{\cos(\varphi) = 0{,}8} \rightarrow \text{Factor de POT}$$

(b) $V_{ef}^2 = V_{ef_R}^2 + V_{ef_L}^2$

$$\Rightarrow V_{ef_L} = \sqrt{V_{ef}^2 - V_{ef_R}^2} = 120\ V =$$

$V_{e_L} = 120\ V$

$V_{e_R} = i_{ef} \cdot R \Rightarrow i_{ef} = 5{,}33\ A$

$$\Rightarrow V_{e_L} = i_e \cdot X_L \Rightarrow X_L = \frac{120\ V}{5{,}33\ A} = 22{,}51\ \Omega$$

$$X_L = \omega L \Rightarrow L = \frac{22{,}51\ \Omega}{500\ s^{-1}} = 0{,}045\ Hy$$